



UCLouvain

LINMA2222 - Stochastic Optimal Control & RL
Project

Optimal Portfolio Strategy

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Question 2.1

The expression of g_t can be seen as a composition of two terms:

$$g_t := 1000 \cdot (q_{t+1}p_{t+1} - q_t p_t - (q_{t+1} - q_t)\bar{p}_{t,t+1})$$

The first term (in blue) represents the change in value between time t and $t + 1$ of our holdings with respect to the market price.

The second term (in red) represents the cash flow from spending money on stocks or selling them at the execution price $\bar{p}_{t,t+1}$. If $q_{t+1} > q_t$, then we bought some stocks and, similarly, if $q_{t+1} < q_t$ then we sold some stocks.

We can thus conclude that g_t represents our (gross) profit/loss between t and $t + 1$.

Question 2.2

The price at the next time is equal to the last time plus 4 factors : $p_{t+1} = p_t + z_t^a + z_t^u + \gamma^u u_t + \sigma^p \xi_t^p$

- z_t^a : This state variable represents the influence of random factors in the evolution of price (e.g. political situation, climate, etc.). It reverts gradually to zero with persistence controlled by ω^a and is driven by normally distributed changes of variance $(\omega^a \sigma^a)^2$. This implies that small values of ω^a create a slow decay of the random part ξ_t^a .
- z_t^u : This state variable is deterministic and represents the temporary effect of the input u_t on the price. It takes the same form as z_t^a with a coefficient ω^u that controls the persistence and a weight β^u to scale the impact of the input. This simulates the impact of your purchase on the market price.
- $\gamma^u u_t$: This term represents the immediate effect of u_t on the next price p_{t+1} . Unlike z_t^u , this term has an effect only on the next time step.
- $\sigma^p \xi_t^p$: This term adds random and immediate fluctuations of the price to account for other phenomena that are not taken into account in the model.

All of these components make sense for an asset's price over time and we have all the major components for a reasonable representation of it. It takes into account external market factors as well as our trade decisions and have a decaying impact and an immediate one.

- $\bar{p}_{t,t+1}$: Finally, the execution price is a weighted average of the price p_t and p_{t+1} using a parameter θ that interpolates both values.

Question 2.3

Let $y_t = (x_t^\top, u_t, \xi_t^\top)^\top = (q_t, z_t^a, z_t^u, u_t, \xi_t^a, \xi_t^p)^\top$. We take the expression of g_t :

$$\begin{aligned}
\frac{g_t}{1000} &= q_{t+1}p_{t+1} - q_t p_t - (q_{t+1} - q_t)\bar{p}_{t,t+1} \\
&= (q_t + u_t)p_{t+1} - q_t p_t - u_t \bar{p}_{t,t+1} \\
&= q_t(p_{t+1} - p_t) + u_t(p_{t+1} - \bar{p}_{t,t+1}) \\
&= q_t(z_t^a + z_t^u + \gamma^u u_t + \sigma^p \xi_t^p) + \theta u_t(p_{t+1} - p_t - \gamma^u u_t) \\
&= q_t(z_t^a + z_t^u + \gamma^u u_t + \sigma^p \xi_t^p) + \theta u_t(z_t^a + z_t^u + \sigma^p \xi_t^p)
\end{aligned}$$

We now have an expression that depends only on the variable y_t and can thus be put in a vector form :

$$\begin{aligned}
\frac{g_t}{1000} &= y_t^\top \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & \frac{\gamma^u}{2} & 0 & \frac{\sigma^p}{2} \\ \frac{1}{2} & 0 & 0 & \frac{\theta}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & \frac{\theta}{2} & 0 & 0 \\ \frac{\gamma^u}{2} & \frac{\theta}{2} & \frac{\theta}{2} & 0 & 0 & \frac{\theta\sigma^p}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sigma^p}{2} & 0 & 0 & \frac{\theta\sigma^p}{2} & 0 & 0 \end{bmatrix} y_t \\
\Rightarrow g_t &= \frac{1}{2} y_t^\top H y_t
\end{aligned}$$

where:

$$H = 1000 \cdot \begin{bmatrix} 0 & 1 & 1 & \gamma^u & 0 & \sigma^p \\ 1 & 0 & 0 & \theta & 0 & 0 \\ 1 & 0 & 0 & \theta & 0 & 0 \\ \gamma^u & \theta & \theta & 0 & 0 & \theta\sigma^p \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \sigma^p & 0 & 0 & \theta\sigma^p & 0 & 0 \end{bmatrix}$$

Question 2.4

As the statement suggests, let's consider that g_t is a random variable. In particular, let's assume it is a uniformly distributed RV. To be zero-mean and of variance σ^2 , g_t must follow this distribution:

$$\begin{aligned}
g_t &\sim \mathcal{U}(-\sqrt{3}\sigma, \sqrt{3}\sigma) \\
\Rightarrow \varphi_{g_t}(g) &= \begin{cases} \frac{1}{2\sqrt{3}\sigma} & \text{if } x \in [-\sqrt{3}\sigma, \sqrt{3}\sigma] \\ 0 & \text{otherwise} \end{cases}
\end{aligned}$$

We can now compute the expectation of the utility function with respect to g_t :

$$\begin{aligned}
\mathbb{E}[c(g_t)] &= \int_{\mathbb{R}} c(g) \varphi_{g_t}(g) \, dg \\
&= \int_{-\sqrt{3}\sigma}^{\sqrt{3}\sigma} \max\left(g - \frac{1}{2}g^2, 1 - \exp(-g)\right) \frac{1}{2\sqrt{3}\sigma} \, dg \\
&= \frac{1}{2\sqrt{3}\sigma} \int_{-\sqrt{3}\sigma}^{\sqrt{3}\sigma} \max\left(g - \frac{1}{2}g^2, 1 - \exp(-g)\right) \, dg \\
&:= F(\sigma)
\end{aligned}$$

On Figure 1, we plotted $F(\sigma)$ for positive values of σ . We observe that the function is indeed decreasing as σ increases.

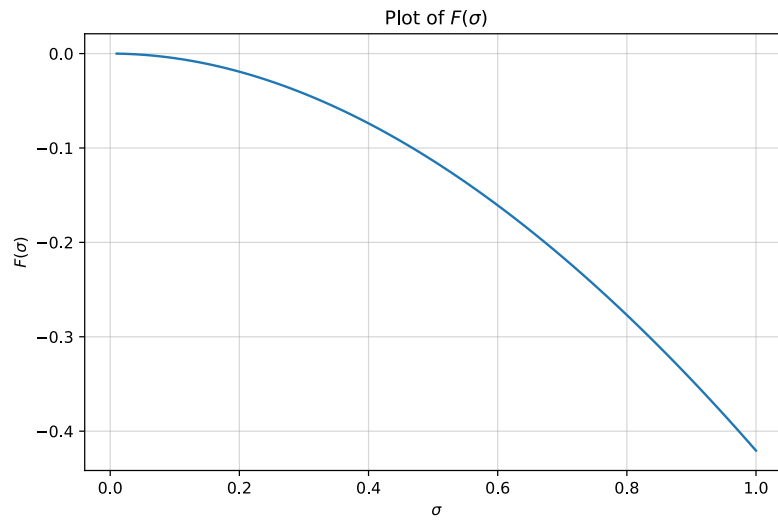


Figure 1: Plot of $F(\sigma)$ for $\sigma > 0$

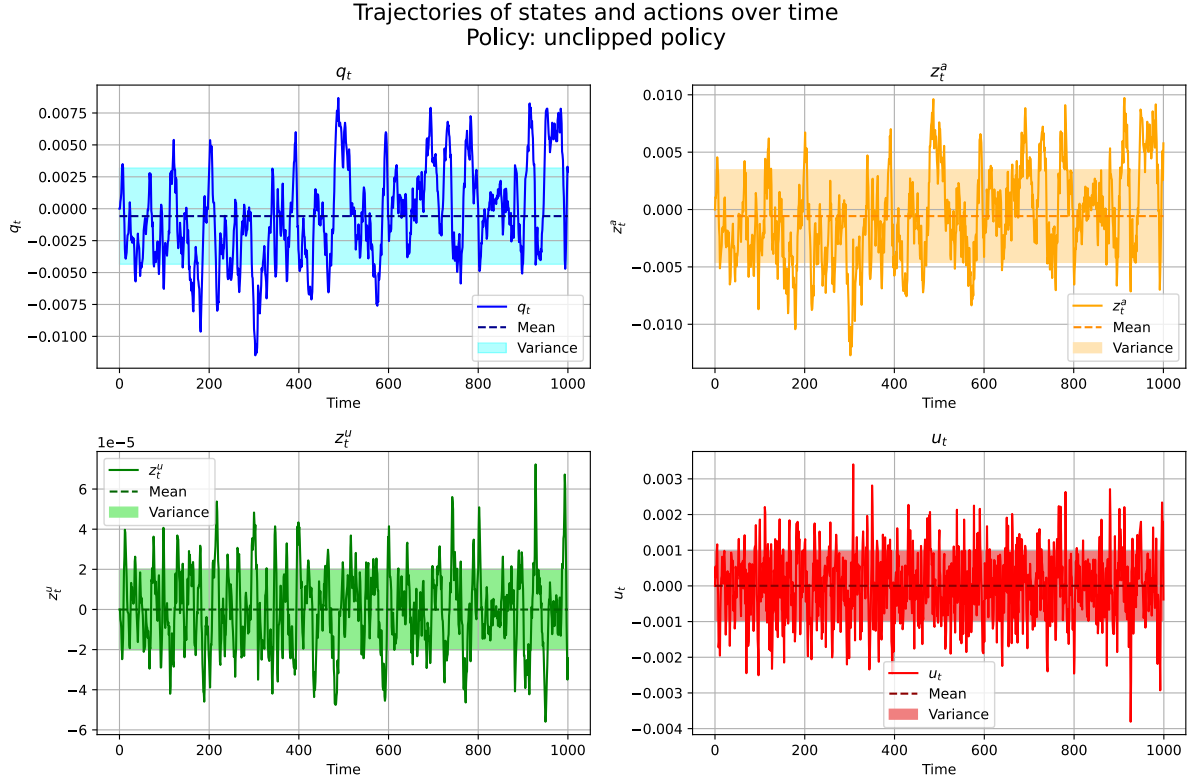
Question 3.1

Let us now simulate the evolution of the dynamical model with the policy π_{cl} defined as:

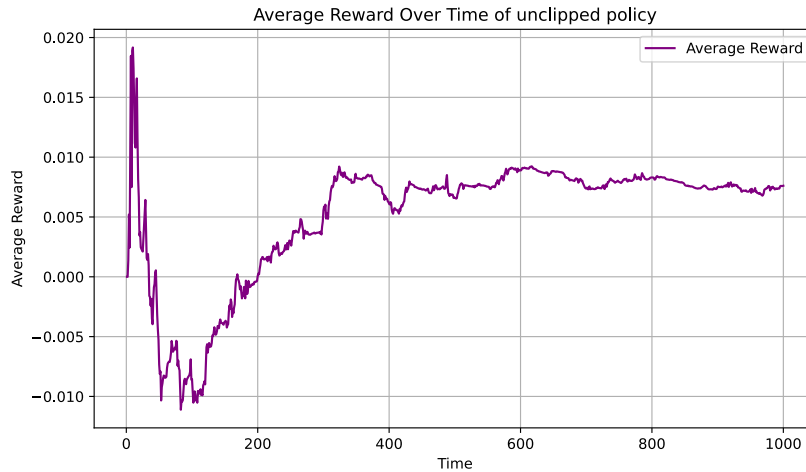
$$\pi_{\text{cl}}(x_t) = K_{\text{cl}}x_t = -0.5q_t + 0.5z_t^a + 0.5z_t^u$$

with a initial condition $x_0 = (0, 0, 0)$ and for time $t = 1, \dots, 1000$.

Here are the plots of the three state variables and the input u_t .



We can observe that the x_t and u_t both seem to be centered around 0. This means that there is as many moments where we buy than moments where we sell assets. Of course, this is not realistic as it results in negative quantities of stocks in our portfolio. This aspect is taken into account in the next strategy where we constrain $q_t \geq 0$.



Here is the average reward between 0 and t for each time t .

We observe that it ends up to a positive constant (≈ 0.0075). This is an indicator that this is a winning policy.

Question 3.2

Let us now run a simulation of the model in the same way but for $N = 100$ trajectories. On Figure 4, we plotted the empirical distribution of the state variables. We can now clearly see that they are all centered around 0. Concerning the variance, we see that q_t and z_t^a seem to have approximately the same variance. Overall, they all seem to follow a normal distribution. z_t^u , however, has a very small variance. This is surely due to the fact that it does not, directly, depends on a random variable.

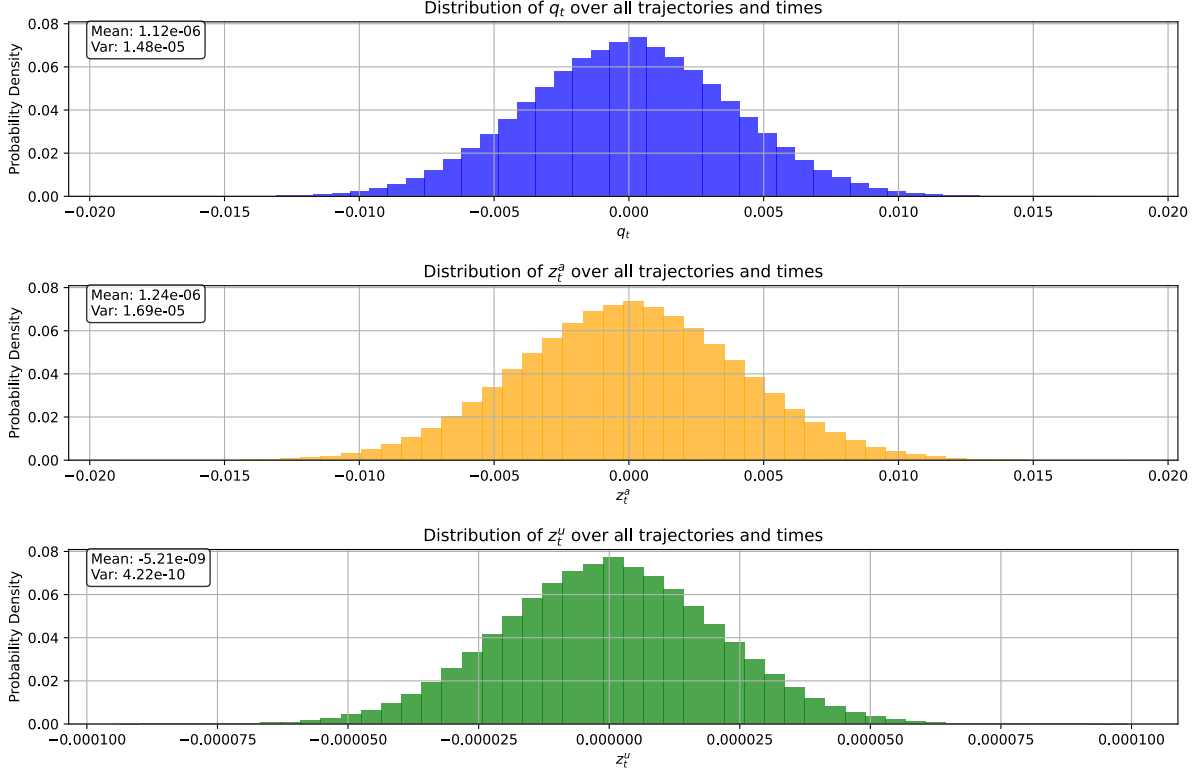


Figure 4: Empirical distribution of the state variables

On Figure 5, we plotted all the distribution of the average reward for every trajectory. We can see at the start, the reward have a high variance and can still be negative but as time goes the average reward tends to a positive $\approx 1.08 \times 10^{-2}$. This means this strategy is profitable in the long term. If we look at the empirical distribution, we see that after 1000 time steps there is practically no chance of losing.

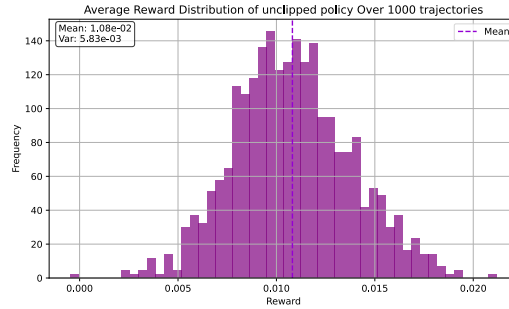


Figure 5: Plot of the average rewards for every trajectory

Question 3.3

The previous policy was not realistic because it could result in negative quantities of assets in our portfolio. To fix this, we consider the following policy: $\pi_{\text{pcl}}(x_t) = \max(-q_t, \min(\pi_{\text{cl}}(x_t)))$ which returns the previously defined policy if it does not break the constraint $q_t \geq 0$, otherwise it clips the returned value between $[-q_t, 1 - q_t]$ which prevents the input to change the value of our portfolio to exceed the range $[0, 1]$.

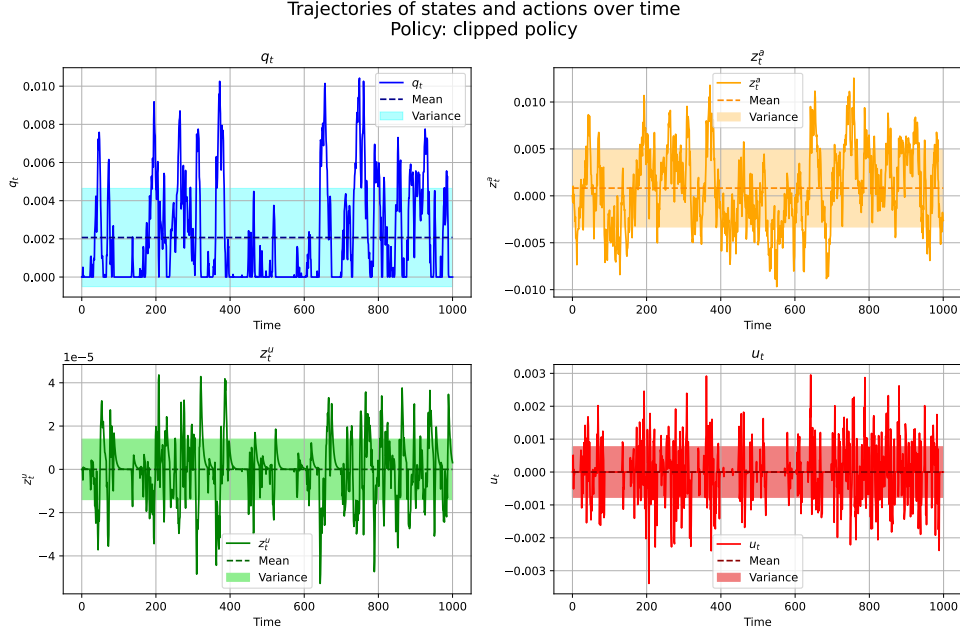
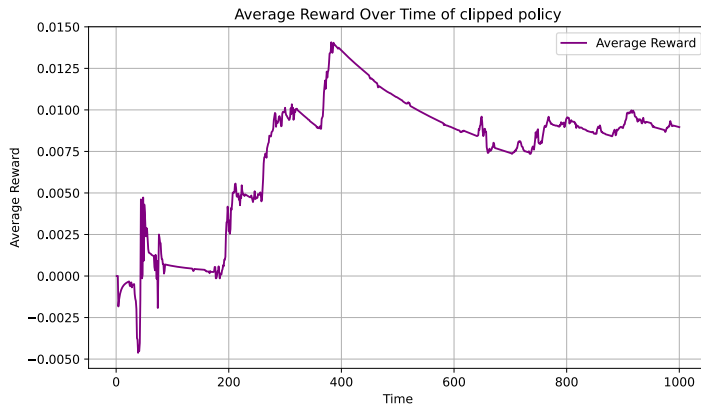


Figure 6: Plots of state variables and input u_t for the clipped policy

On Figure 6, we first notice that q_t indeed satisfies the constrained range. We can also notice some plateaux for some regions for q_t and u_t . Those plateaux represent moments where we previously wanted to reach a negative quantity q_t in our portfolio but the policy won't allow it. It thus result in a clipped value to 0. For those regions, we also observe that z_t^u decays to 0 because the input is null.



What is more interesting is that we observe multiple drops in the average reward over time and that those drops correspond exactly to the time the plateaux occur. An explanation for this would be that the unclipped policy would have been able to leverage the loss – or even win – but the constraint prevents those actions from being made which results in unwanted choice from our policy. Of course, as a consequence of this, we observe that the long term average reward seems to tend to a lower value.

Question 3.4

We ran the simulation again for $N = 100$ trajectories with the clipped policy. On Figure 8, we clearly see that q_t is always non-negative. Consequently, its mean is greater than for the previous policy. It also results in the probability for q_t to be equal to 0 to be very high, as all previously negative values are now projected to 0. The same goes for z_t^u where we can clearly see the previously mentioned decays due to the input u_t being null.

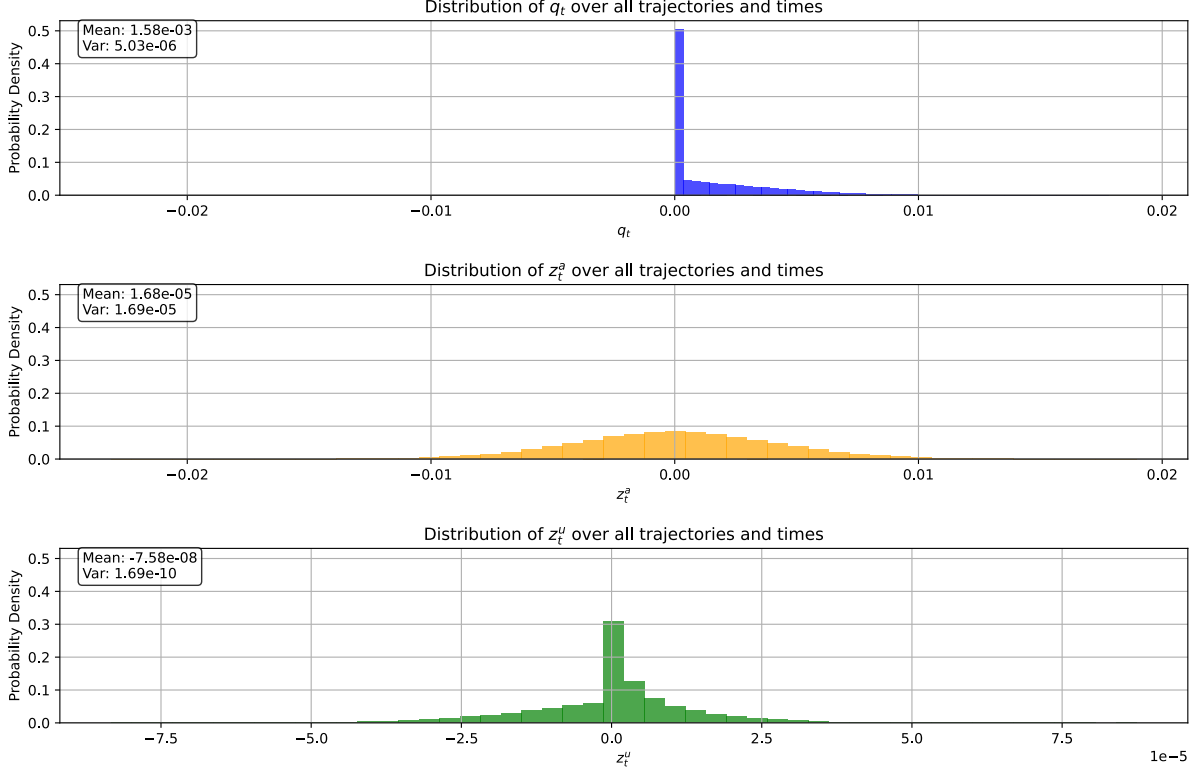


Figure 8: Empirical distribution of the state variables

On Figure 9, we finally plotted the reward distribution like before. The variance at steady-state seems to be of the same magnitude as for the previous policy. However, as mentioned in the previous section, we observe that the infinite horizon average reward seems to have a smaller mean. This is because, while this policy is more realistic, it does not permit to make “illegal” decisions which allowed us to make more profit.

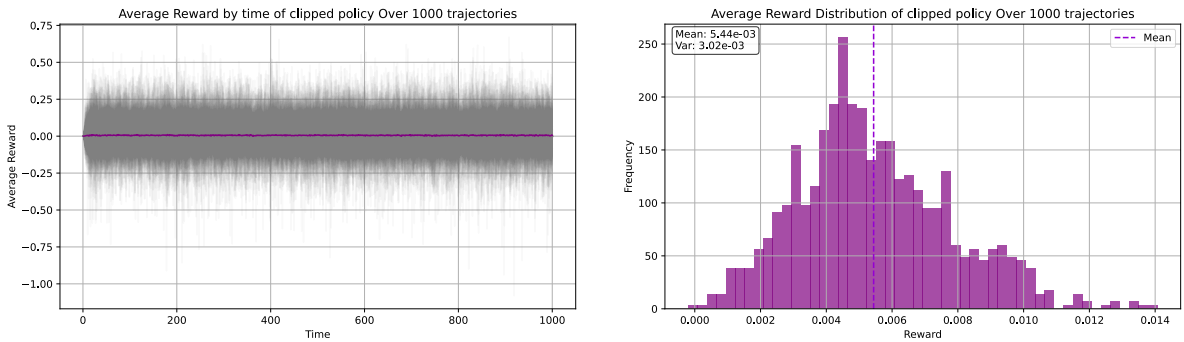


Figure 9: Plot of the average rewards and the distribution at the final time step

Question 3.5

To find a better policy, we will optimize the linear policy :

$$\pi_l^*(x_t) = \max(q_t, \min(1 - q_t, p_3[q_t, z_t^a, z_t^u]^\top))$$

where $p_3 = [\alpha, \beta, \gamma]$ is the array of 3 parameters. Of course, we clip the value of the output to satisfy the constraints.

We optimized the reward at final time step $T = 1000$ averaged on $N = 1000$ simulations with CMA-ES solver from the `cma` library. This solver is made for stochastic function as opposed to a solver like `SciPy` and works way better.

The solver ran tests 20.000 times at $N = T = 1000$ for about 20 minutes and found $p_3 = [-0.0529, 37.2788, 29.9648]$

The linear optimal policy is a bit limited. That's why we added quadratic relations so it can only do better :

$$\pi_q^*(x_t) = \max(q_t, \min(1 - q_t, p_{10}[q_t, z_t^a, z_t^u, (q_t)^2, (z_t^a)^2, (z_t^u)^2, q_t z_t^a, z_t^a z_t^u, z_t^u q_t, 1]^\top))$$

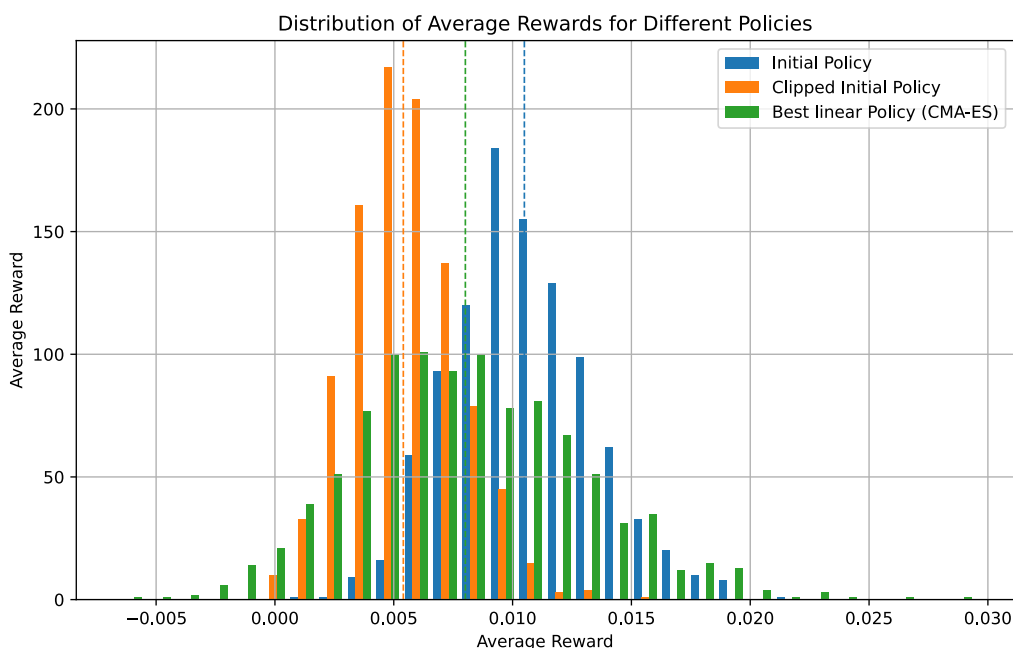
where p_{10} , the 10 parameters to optimize this quadratic policy.

$p_{10} = [-0.1674, 25.6970, 7.1492, 0.0350, 17.5269, -4.0057, 9.3930, 24.3607, -0.5704, 0.0603]$

Comparison

To compare all the policies infinite-horizon average reward, we plotted their distributions. We can clearly see how our policies performs way better than the previous two, even though there is a larger variance in the average reward.

The quadratic policy performed marginally better than the linear policy. Let's analyze this quadratic policy.



Question 3.6

Again, let us plot the states variables, the input, as well as the average reward for a single trajectory.

On Figure 11, we first notice that q_t satisfies the constraint at all time steps. We also notice that q_t reaches extreme values more often than for the other policies. This seems pretty logic since we logically want to buy all the stocks available if there is a good opportunity, and we want to sell all of our stocks if we predict that the price will go down.

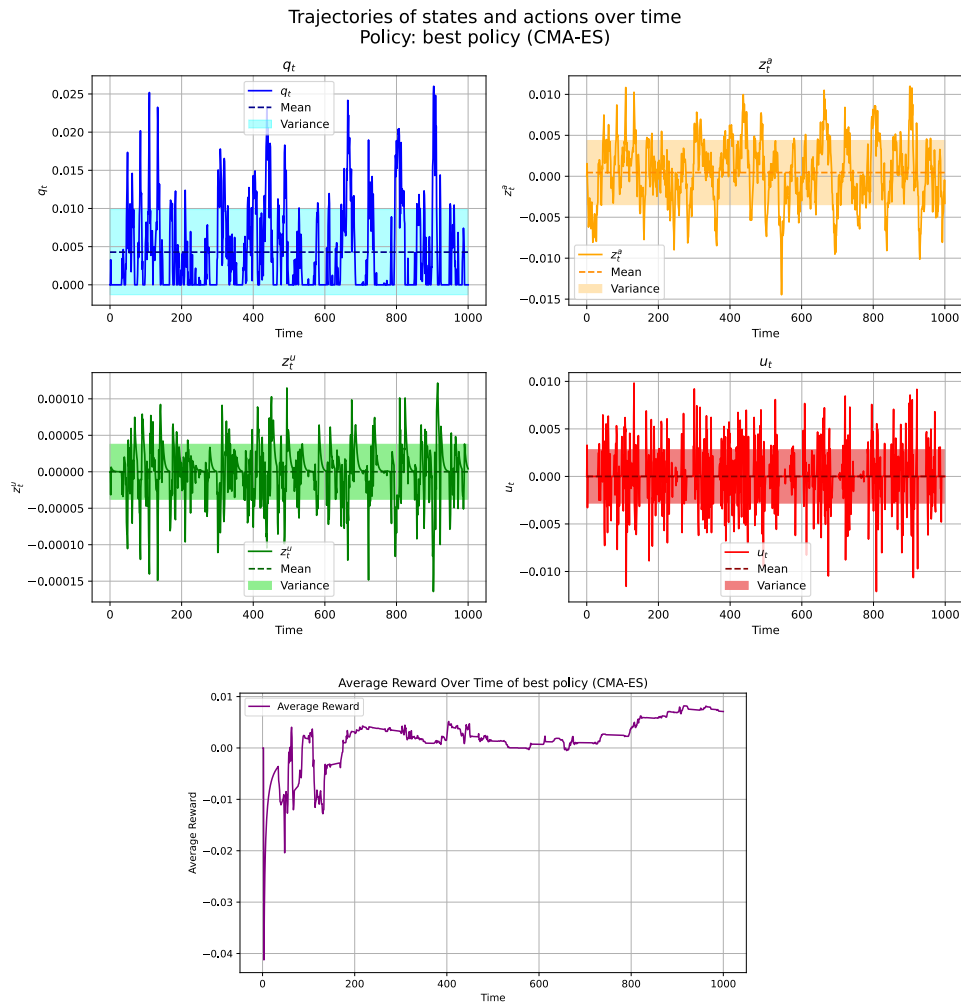


Figure 11: Plots of state variables, input u_t and average reward for our policy

Question 3.7

Now, let's run 100 trajectories and take a look at the empirical distributions as before. On Figure 12, we clearly observe the phenomenon mentioned above where the values of q_t are mainly 0 or 1.

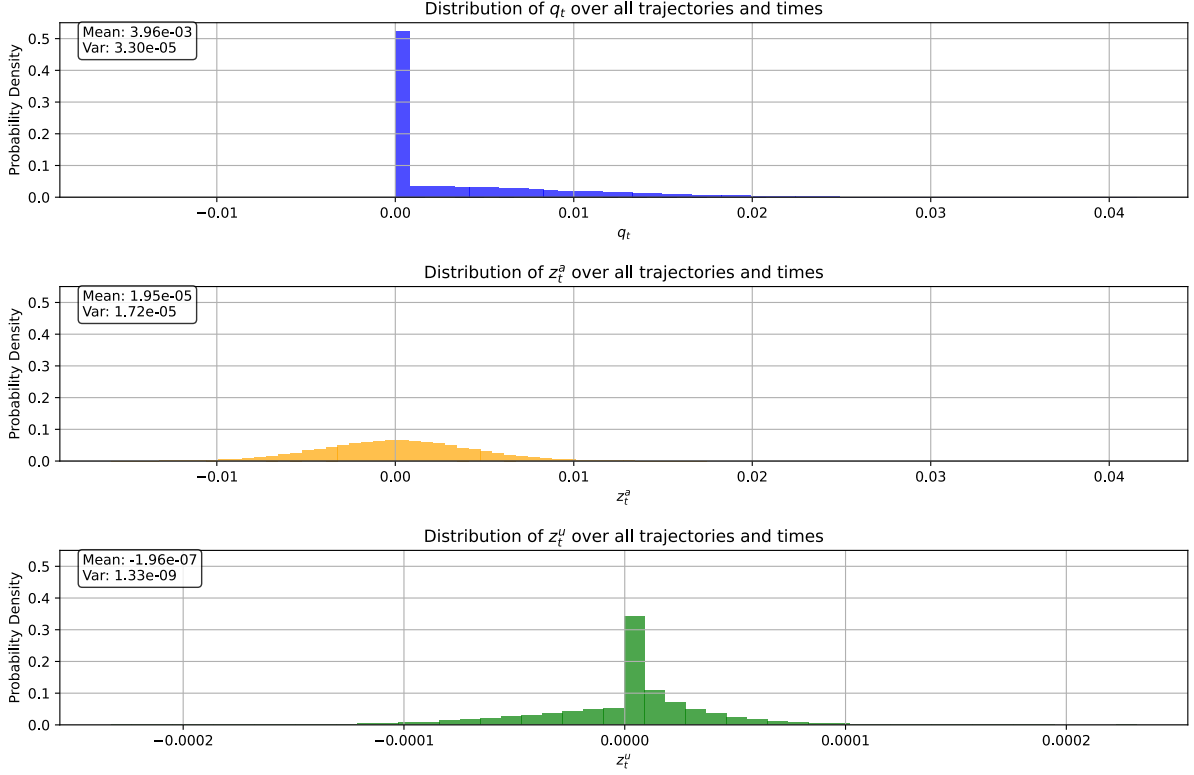


Figure 12: Empirical distribution of the state variables

What is more interesting is that we observe a way better infinite-horizon average reward for this policy ($\approx 7 \times 10^{-4}$). It is 10 times higher than the first, non clipped policy.

The drawback is that there is now a higher probability to get a negative reward after $T = 1000$ time steps.

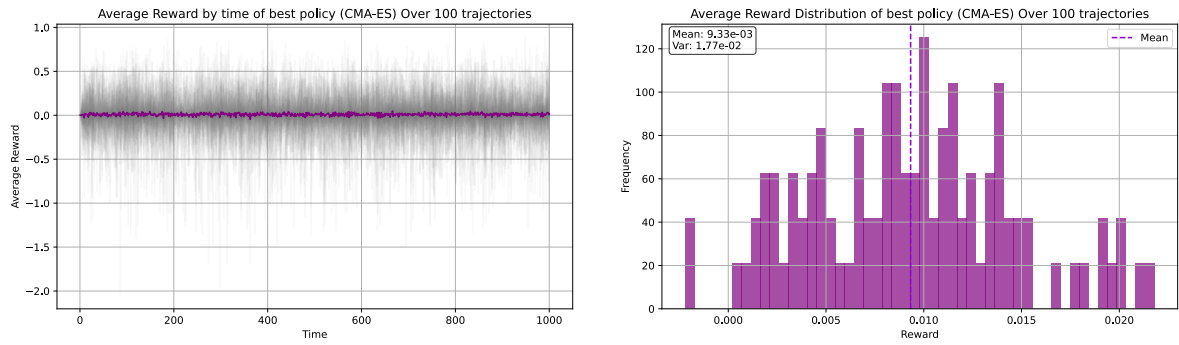


Figure 13: Plot of the average rewards and the distribution at the final time step

Question 4.1

We know the following :

$$\begin{aligned} q_{t+1} &= q_t + u_t \\ z_{t+1}^a &= (1 - \omega^a)z_t^a + \omega^a \sigma^a \xi_t^a \\ z_{t+1}^u &= (1 - \omega^u)z_t^u + \omega^u \beta^u u_t \end{aligned}$$

Therefore, we can express the explicit linear form : $x_{t+1} = Fx_t + Gu_t + D\xi_t$,

$$\text{with } F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1-\omega^a & 0 \\ 0 & 0 & 1-\omega^u \end{bmatrix}, G = \begin{bmatrix} 1 \\ 0 \\ \omega^u \beta^u \end{bmatrix}, D = \begin{bmatrix} 0 & 0 \\ \omega^a \sigma^a & 0 \\ 0 & 0 \end{bmatrix}$$

Question 4.2

Let $L(x_t, u_t, \xi_t) := g_t$ be the function representing g_t . We have :

$$c_{\text{quad}}(g_t) = L(x_t, u_t, \xi_t) - \frac{1}{2}L(x_t, u_t, \xi_t)^2$$

Taking the conditional expectation $\mathbb{E}[\cdot \mid x_t, u_t]$:

$$\mathbb{E}[c_{\text{quad}}(g_t) \mid x_t, u_t] = \mathbb{E}[L(x_t, u_t, \xi_t) \mid x_t, u_t] - \frac{1}{2}\mathbb{E}[L(x_t, u_t, \xi_t)^2 \mid x_t, u_t]$$

Since ξ_t is zero-mean and is independent of x_t, u_t , we have:

$$\begin{aligned} \mathbb{E}[L(x_t, u_t, \xi_t) \mid x_t, u_t] &= L(x_t, u_t, 0) \quad (\text{deterministic part}) \\ \mathbb{E}[L(x_t, u_t, \xi_t)^2 \mid x_t, u_t] &= L(x_t, u_t, 0)^2 + \mathbb{V}[L(x_t, u_t, \xi_t) \mid x_t, u_t] \end{aligned}$$

So:

$$\begin{aligned} \mathbb{E}[c_{\text{quad}}(g_t) \mid x_t, u_t] &= L(x_t, u_t, 0) - \frac{1}{2}L(x_t, u_t, 0)^2 - \frac{1}{2}\mathbb{V}[L(x_t, u_t, \xi_t) \mid x_t, u_t] \\ &\approx L(x_t, u_t, 0) - \frac{1}{2}\mathbb{V}[L(x_t, u_t, \xi_t) \mid x_t, u_t] \end{aligned}$$

We want to keep only the terms of second order. We thus approximated the previous expression by omitting $L(x_t, u_t, 0)^2$. Let's now identify $L(x_t, u_t, 0)$ and the variance.

From question 2.3, we have:

$$\begin{aligned} L(x_t, u_t, 0) &= 1000(q_t z_t^a + q_t z_t^u + \theta u_t z_t^a + \theta u_t z_t^u + q_t \gamma^u u_t) \\ &= x_t^\top \begin{bmatrix} 0 & 500 & 500 \\ 500 & 0 & 0 \\ 500 & 0 & 0 \end{bmatrix} x_t + x_t^\top \begin{bmatrix} 1000 \cdot \gamma^u \\ 1000 \cdot \theta \\ 1000 \cdot \theta \end{bmatrix} u_t \end{aligned}$$

For the variance, the only stochastic part of g_t is:

$$1000(q_t \sigma^p \xi_t^p + u_t \theta \sigma^p \xi_t^p) = 1000 \sigma^p (q_t + \theta u_t) \xi_t^p$$

So:

$$\begin{aligned}
\mathbb{V}[L(x_t, u_t, \xi_t) \mid x_t, u_t] &= (1000\sigma^p(q_t + \theta u_t))^2 \mathbb{V}[\xi_t^p] = (1000\sigma^p)^2 (q_t + \theta u_t)^2 \\
&= (1000\sigma^p)^2 (q_t^2 + 2q_t\theta u_t + \theta^2 u_t^2) \\
&= x_t^\top \begin{bmatrix} (1000\sigma^p)^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} x_t + x_t^\top \begin{bmatrix} 2\theta(1000\sigma^p)^2 \\ 0 \\ 0 \end{bmatrix} u_t + (1000\sigma^p)^2 \theta^2 u_t^2
\end{aligned}$$

Combining these two expressions, we obtain:

$$\begin{aligned}
\mathbb{E}[c_{\text{quad}}(g_t) \mid x_t, u_t] &\approx x_t^\top \begin{bmatrix} -\frac{1}{2}(1000\sigma^p)^2 & 500 & 500 \\ 500 & 0 & 0 \\ 500 & 0 & 0 \end{bmatrix} x_t + x_t^\top \begin{bmatrix} 1000 \cdot \gamma^u - \theta(1000\sigma^p)^2 \\ 1000 \cdot \theta \\ 1000 \cdot \theta \end{bmatrix} u_t - \frac{1}{2}(1000\sigma^p)^2 \theta^2 u_t^2 \\
&= \frac{1}{2} x_t^\top S x_t + x_t^\top P u_t + \frac{1}{2} u_t^\top R u_t
\end{aligned}$$

with:

$$S = \begin{bmatrix} -(1000\sigma^p)^2 & 1000 & 1000 \\ 1000 & 0 & 0 \\ 1000 & 0 & 0 \end{bmatrix}, \quad P = \begin{bmatrix} 1000 \cdot \gamma^u - \theta(1000\sigma^p)^2 \\ 1000 \cdot \theta \\ 1000 \cdot \theta \end{bmatrix}, \quad R = [-(1000\sigma^p)^2 \theta^2]$$

Question 4.4

For the LQR problem with linear dynamics and quadratic reward, the optimal policy that maximizes the infinite-horizon average reward is identical to the optimal policy that maximizes the infinite-horizon total reward, which is also identical to the policy that minimizes the infinite-horizon total cost (where cost = negative reward). Therefore, we can use the standard LQR solution procedures seen in class by converting our reward maximization into an equivalent cost minimization problem. We thus define:

$$J(u) = \min_u \mathbb{E} \left[\sum_{t=0}^{\infty} c(x_t, u_t) \right]$$

where $c(x_t, u_t) := -\hat{r}_t(x_t, u_t)$. Taking the expectation on the reward defined in the statement is equivalent to considering the dynamical model without the noise because the reward is linear in terms of x_t and that x_t is itself linear in terms of ξ_t which is zero-mean. We thus obtain:

$$J(u) = \min_u \sum_{t=0}^{\infty} c(\bar{x}_t, u_t)$$

where $\bar{x}_{t+1} = F\bar{x}_t + Gu_t$ is the same dynamical model as before but without the noise. We finally obtained the same formulation as the LQR problem seen in course. [1] By defining:

$$Q := -\frac{1}{2}S, \quad \hat{R} := -\frac{1}{2}R, \quad N := -\frac{1}{2}P$$

We find the optimal policy to be:

$$u_k = -Kx_k$$

$$K = \left(\hat{R} + G^\top M^* G \right)^{-1} (G^\top M^* F + N^\top)$$

where M^* is the unique solution of the Ricatti equation:

$$M^* = F^\top M^* F - (F^\top M^* G + N) \left(\hat{R} + G^\top M^* G \right)^{-1} (G^\top M^* F + N^\top) + Q$$

Which we will solve by iterating on the finite horizon Ricatti equation until M^* converges. After solving these equations, we obtained the following optimal gain:

$$K \approx [1.112 \quad -2.649 \quad -2.528]$$

$$\pi_{\text{lqr}}(x_t) = -Kx_t = -1.112q_t + 2.649z_t^a + -2.528z_t^u$$

To compute the average approximate reward theoretically, we used the following formula:

$$J_{\text{reward}}^* = -\frac{1}{2} \text{tr}(M^* D D^\top) \approx 0.01936$$

Question 4.5

We will now recreate the plots from Question 3.1 with the LQR policy.

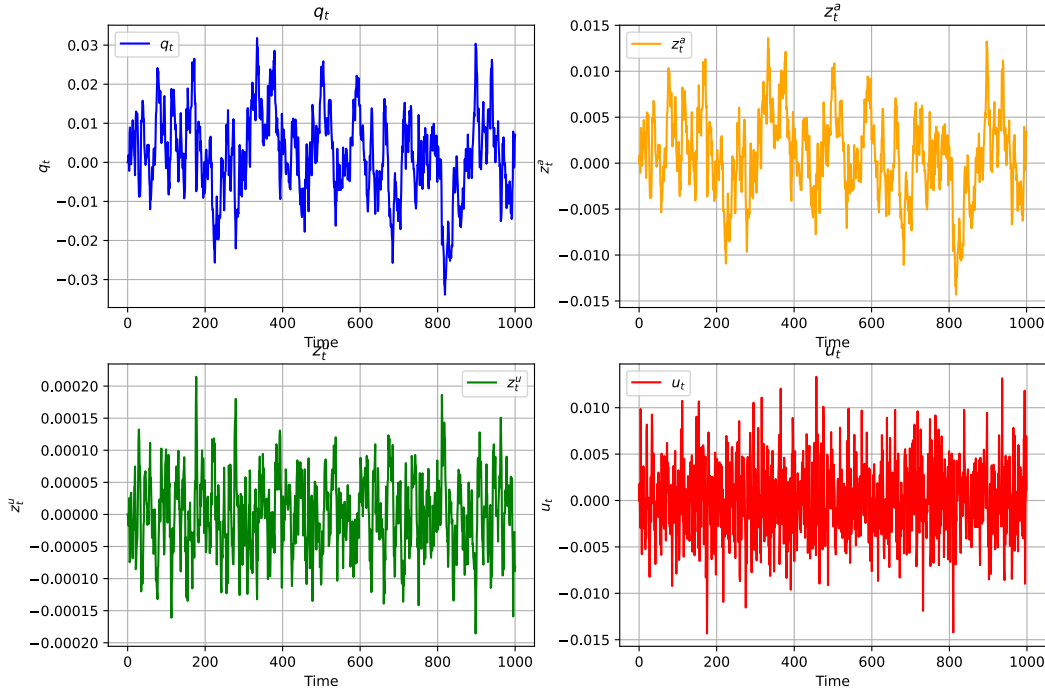


Figure 14: Plots of state variables and input u_t for the LQR policy

Compared to the initial unclipped policy π_{cl} , we observe on Figure 14 that the quantities in stock take bigger values (0.0075 compared to 0.03, so about 4 times more). The same goes for the input u_t where we observe about 2 to 3 times higher values. This means that the LQR policy is more aggressive. However, by being more aggressive at the right time, the LQR policy achieves a much greater average reward :

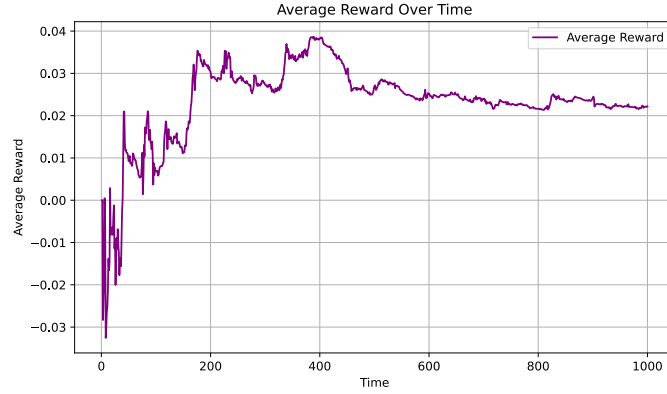


Figure 15: Plots of state variables and input u_t for the LQR policy

We observe on Figure 15 an average reward at the final time step of ≈ 0.0222 , which is very close to the theoretical average reward we computed in the previous section. Compared to the initial policy, this represents an increase of order 10 in the average reward.

Question 4.6

Let's now plot the empirical distributions of the state variables as we did in Question 3.2. Apart from the variance of q_t which is a bit higher than for the initial policy, there are no clear differences between the state distributions of the initial policy and the LQR optimal policy as we can see on Figure 16.

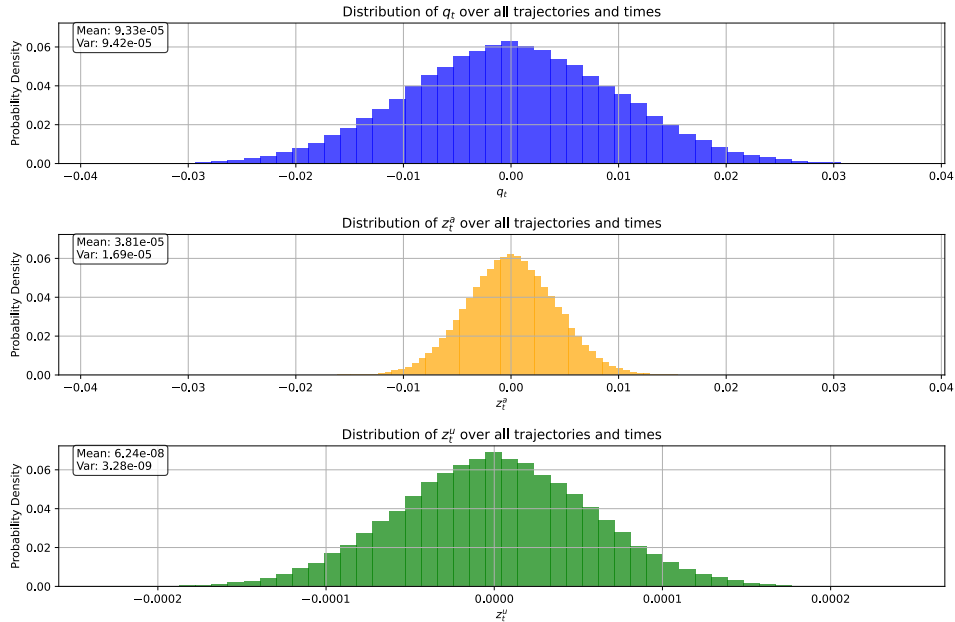


Figure 16: Empirical distribution of the state variables

On Figure 17, we clearly see that the average reward has a much higher variance than with the initial policy (5.83×10^{-3} vs. 3.54×10^{-2}). However, despite this huge variance, this policy still performs better. On average, the average reward has a value of 1.68×10^{-2} , which again is very close to the theoretical value. This indicates that our LQR policy works as intended.

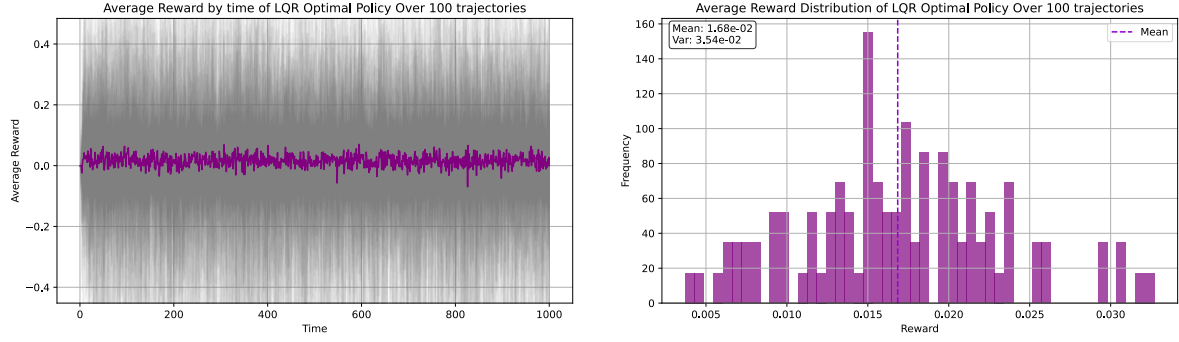


Figure 17: Plot of the average rewards for every trajectory

Question 4.7

Like π_{cl} , the LQR policy is unrealistic because it allows to possess negative quantities of stocks in our portfolio. To prevent this, we clipped the policy so that $q_t \in [0, 1]$. We will call this new clipped policy π_{plqr} .

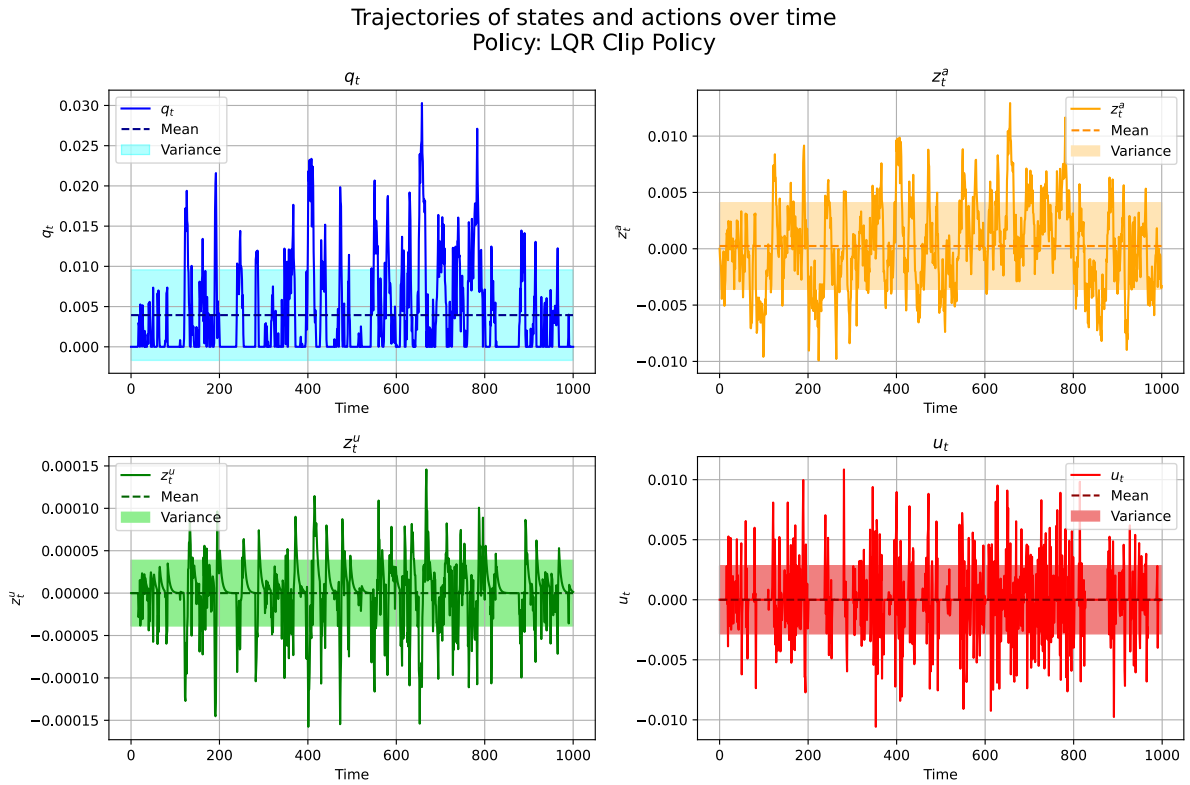


Figure 18: Plots of state variables and input u_t for the LQR clipped policy

The same observation that we noticed in the graphs of the π_{pcl} policy's states can be seen on Figure 18. We thus expect the average reward to be worse than the unclipped LQR policy.

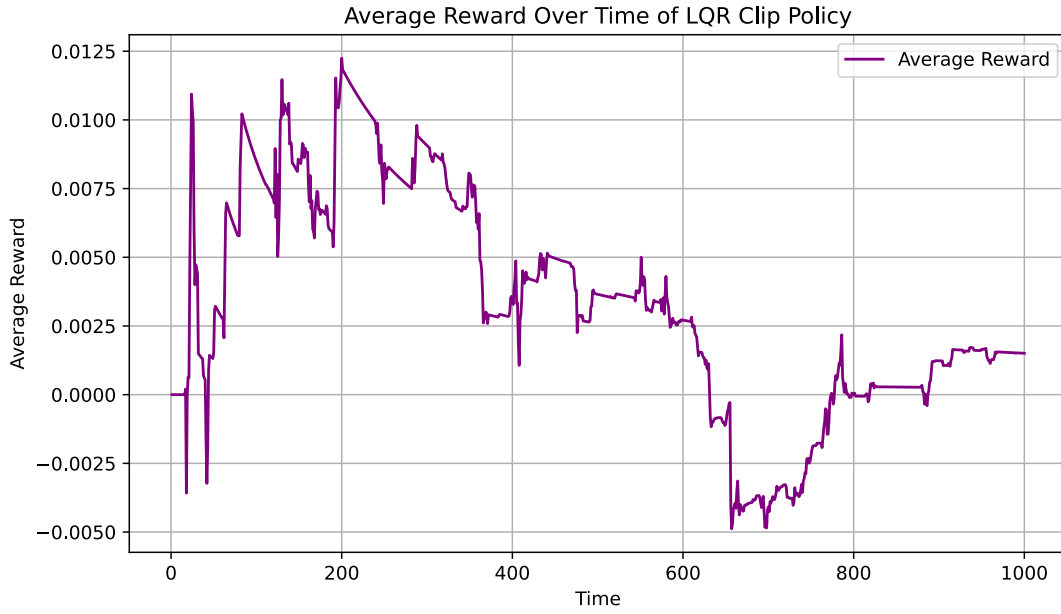


Figure 19: Plots of state variables and input u_t for the LQR clipped policy

For the particular simulation on Figure 19, we observe an average reward that is way worse than before, with a final value of ≈ 0.002 . Though to get a real comparison, we should look at the empirical distributions of the average reward.

Question 4.8

As before, we plotted the average reward for 100 simulations. On Figure 20, we clearly see the huge variance just like the unclipped LQR policy. However, we see that the average reward at the final time step is on average much worse than the unclipped LQR (1.68×10^{-2} vs. 8.86×10^{-3}). As we said previously, this is totally expected as the clipped policy can not make all the “illegal” moves it wished to perform.

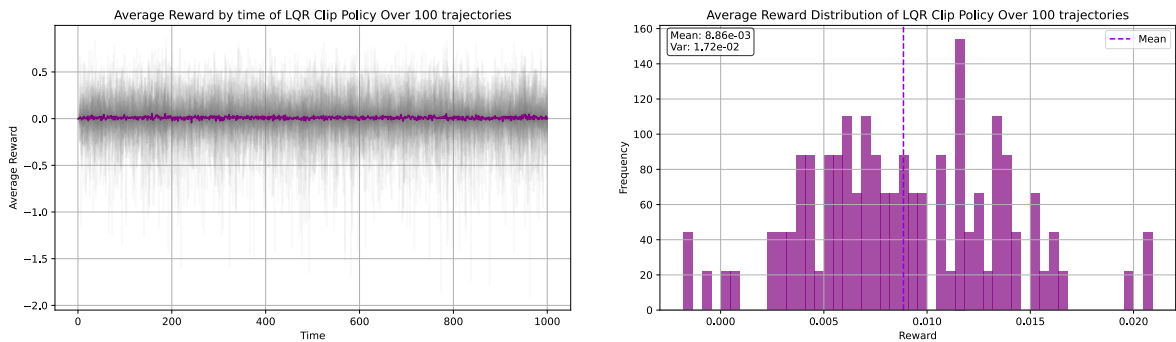


Figure 20: Plot of the average rewards for every trajectory

Question 4.9

Obviously, choosing a bigger horizon $\mathcal{N}$ comes at a cost. We thus have to make a compromise between the computational cost and the reward. After testing multiple values of $\mathcal{N}$ ranging

from 1 to 15, we saw that a value of $\mathcal{N} = 10$ gave pretty good results and was not excessively expensive to compute.

Question 4.10

As before, we plotted the trajectories for the MPC policy with $\mathcal{N} = 10$.

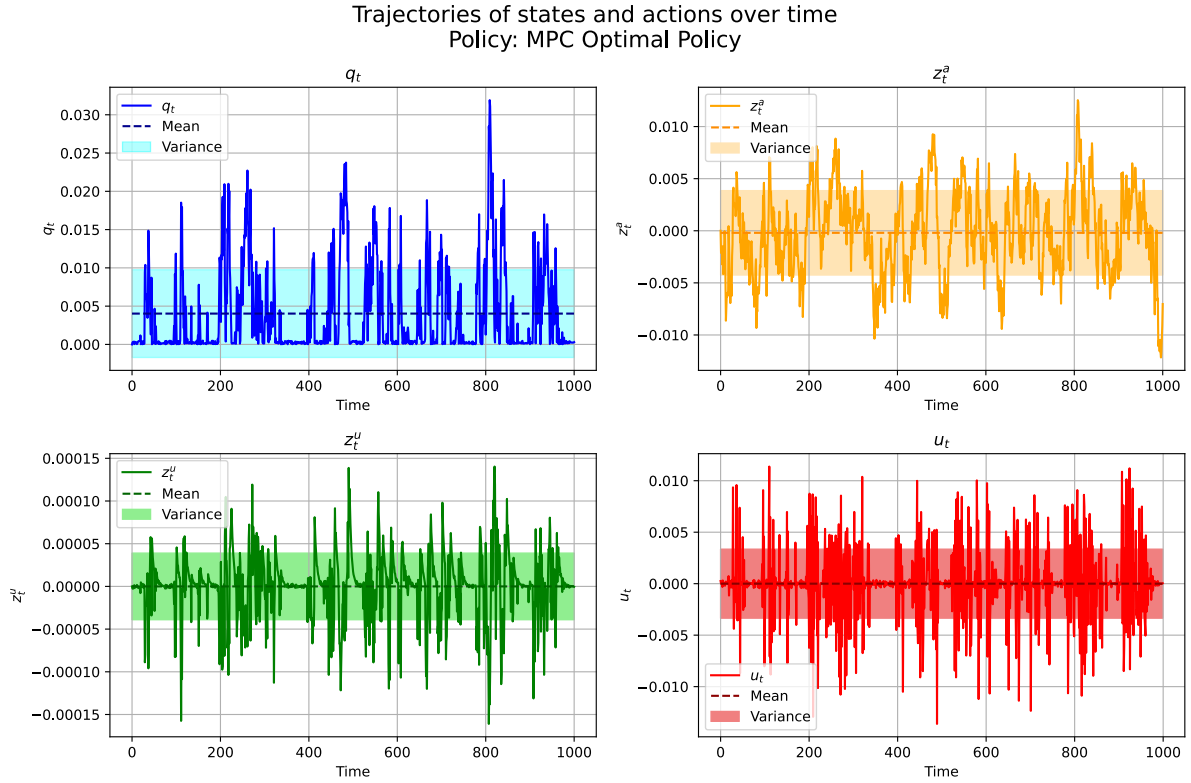


Figure 21: Plots of state variables and input u_t for the MPC policy

The trajectories look quite similar to those of the clipped LQR policy, at least on this particular run. We expect the MPC to have the same performance as the latter.

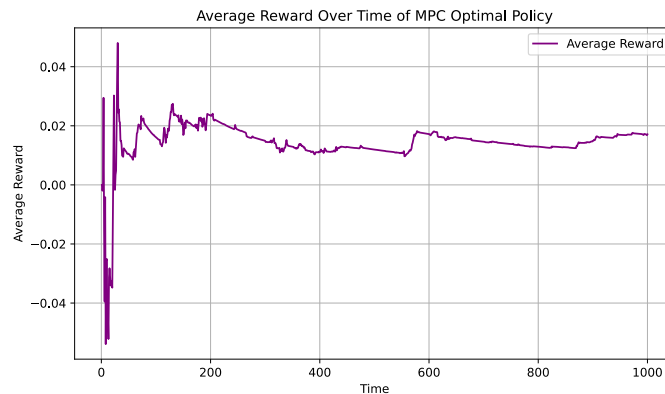


Figure 22: Plots of state variables and input u_t for the MPC policy

On Figure 22, we observe that the average reward at the final time step is ≈ 0.018 , which is quite good and better than what we obtained for the previous policy. Let's now further analyze this by looking at the average performance.

Question 4.11

To look at the average performance, we ran 100 simulations like before to plot the empirical distribution of the average reward. We observe on Figure 23 that we have an average of $\approx 8.49 \times 10^{-3}$. The MPC thus has a performance similar to the clipped LQR policy.

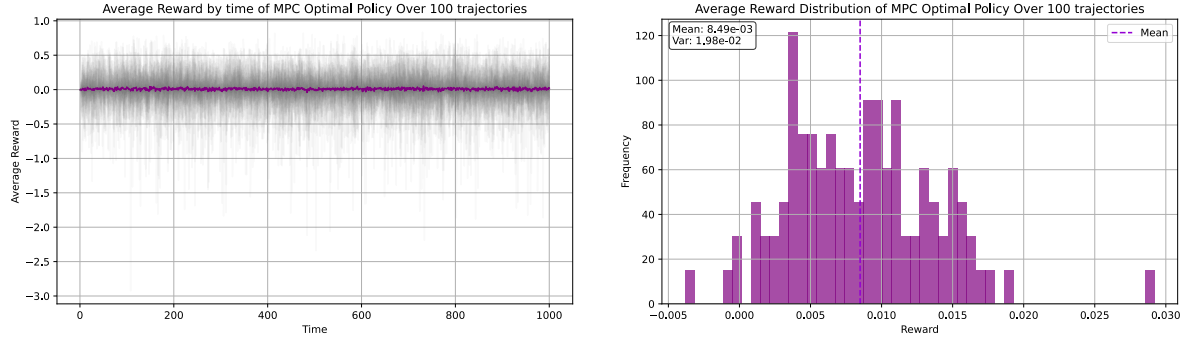


Figure 23: Plot of the average rewards for every trajectory

Conclusion

Based on the simulations, we can deduce that the best policy is the LQR optimal policy. However, due to the fact that it breaks the constraint $q_t \in [0, 1]$, one may want to choose between MPC and the clipped version of the LQR policy.

Because MPC requires to solve an optimization problem at each step and that it, experimentally, did not show significant improvements compared to the LQR, we think that picking the clipped LQR would be a more suited choice for this application.

Question 5.1

We implemented E-PIA with the following notations :

Minimize the infinite-horizon total cost: $J(K) = \sum_{t=0}^{\infty} (x_t^T Q x_t + 2x_t^T S u_t + u_t^T R u_t)$
 with $x_{t+1} = Fx_t + Gu_t$ $u_t = Kx_t$

E-PIA : start with $K_0 = K_{cl}$

1. Compute the equivalent cost and closed-loop matrix

$$F_k = F + GK_k \quad Q_k = Q + SK_k + K_k^T S^T + K_k^T R K_k$$

and find P_k by solving $P_k = Q_k + A_k^T P_k A_k$

2. Compute : $K_{k+1} = -(R + G^T P_k G)^{-1} (S^T + G^T P_k F)$
3. if $\|K_{k+1} - K_k\|_F < \varepsilon$, then stop and set $K^* = K_{k+1}$.

Otherwise, increment $k := k + 1$ and repeat.

We ran it and on the plot shown below, we observe the gain converges to the LQR gain K_{lqr} .

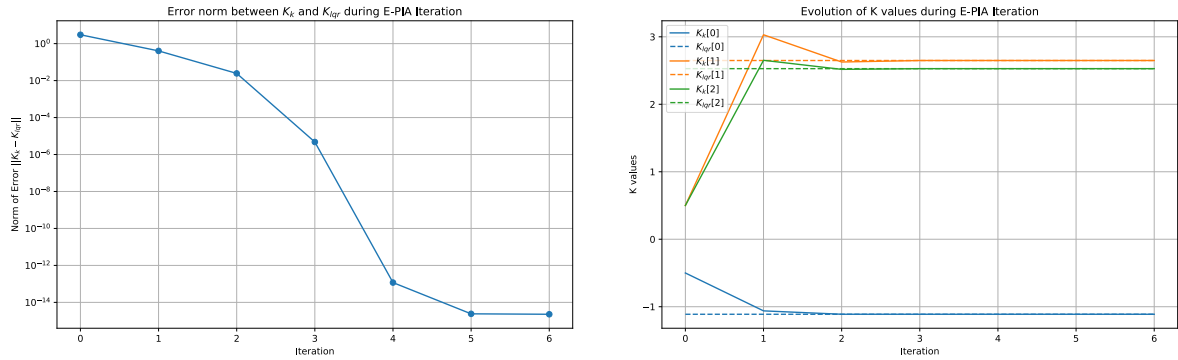


Figure 24: E-PIA Gain K_k evolution compared to LQR gain

Question 6.1

We want to find θ_N^{LSTD} such that $\Gamma_N(\theta) = \theta^\top W \theta + \sum_{k=0}^{N-1} [\gamma_k - \Upsilon_{k+1}^\top \theta]^2$ is minimal. This is a quadratic function so we simply have to impose the first order optimality condition $\nabla_\theta \Gamma(\theta) = 0$:

$$\begin{aligned}
\nabla_{\theta} \Gamma(\theta) &= 2W\theta + 2 \sum_{k=0}^{N-1} \Upsilon_{k+1} [\Upsilon_{k+1}^{\top} \theta - \gamma_k] = 0 \\
&\Leftrightarrow \left[W + \sum_{k=0}^{N-1} \Upsilon_{k+1} \Upsilon_{k+1}^{\top} \right] \theta = \sum_{k=0}^{N-1} \Upsilon_{k+1} \gamma_k \\
&\stackrel{\cdot N-1}{\Leftrightarrow} \left[\frac{1}{N} W + \frac{1}{N} \sum_{k=0}^{N-1} \Upsilon_{k+1} \Upsilon_{k+1}^{\top} \right] \theta = \frac{1}{N} \sum_{k=0}^{N-1} \Upsilon_{k+1} \gamma_k \\
&\Leftrightarrow \theta_N^{\text{LSTD}} = \left[\frac{1}{N} W + \underbrace{\frac{1}{N} \sum_{k=1}^N \Upsilon_k \Upsilon_k^{\top}}_{R_N} \right]^{-1} \underbrace{\left[\frac{1}{N} \sum_{k=0}^{N-1} \Upsilon_{k+1} \gamma_k \right]}_{\bar{\psi}_N^{\gamma_N}}
\end{aligned}$$

Question 6.2

In the course, we saw multiple ways to choose the approximation basis ψ .

Gaussian kernels

One way to approximate the Q -function is by choosing a basis in a large function space called kernels. One choice of kernels is:

$$\mathbb{k}(z, z_i) = \exp\left(-\frac{\|z - z_i\|^2}{2\sigma^2}\right)$$

where we take $z_i = (x_i, u_i)^{\top}$. In the case of a quadratic loss and a quadratic regularizer, the reference book also gives a simple formula for the optimal coefficients (cfr. Meyn pg.168 eq. 5.18 [2]). In the case of a small dataset (i.e. N not too large), this choice of basis may look interesting for the policy improvement step as we have a closed form expression for the optimal coefficients. However, we have a basis of dimension N . For large datasets, this step is computationally expensive because we have to invert a matrix of size $N \times N$.

Polynomial basis

A more straightforward choice would be to simply take the set of functions composed of monomials up to a certain degree. This has the advantage of offering a really cheap policy function evaluation-wise. Let's discuss the choice of the maximum degree for our polynomes.

For a maximum degree of 1, we obtain linear polynomials in term of the state x and the input u . We thus have the following basis:

$$\psi(x, u) = (1, q, z^a, z^u, u)^{\top}$$

This choice is too simple and thus can not capture any non-linearities such as a quadratic structure. To achieve that, let's go up to a degree of 2. We obtain the following basis:

$$\psi(x, u) = (1, q, z^a, z^u, u, q^2, (z^a)^2, (z^u)^2, u^2, qz^a, qz^u, qu, z^a z^u, z^a u, z^u u)^{\top}$$

We have $d = 15$ parameters to find, which is still reasonable. This basis allows to fit exactly a quadratic Q -function such as the one in the LQR problem. More precisely, we have to take only the terms of order exactly 2 to perfectly fit the Q -function in the LQR case.

What if we want to capture even more complex functions ? Unfortunately, increasing the degree of the basis makes the number of parameters increase drastically. For a degree = 3, we find ourselves with a basis of size 35.

In our case, if we suppose that our system is linear and our objective is quadratic, we fall in the LQR case. We know that the Q -function is thus quadratic. Choosing a polynomial basis with a degree of 2 is obviously the best choice.

Question 6.3

In this question, we are asked to implement the LSTD algorithm to obtain an approximation of the Q -function for the π_{cl} policy. To do that, we first need to generate a dataset on the dynamical model where we removed the stochastic parts using the π_{exp} policy given in the statement. We also need to start with an initial condition. In our case, we chose $x_0 = (1, 1, 1)^\top$.

Concerning the choice of basis, our cost (and thus our Q -function) is either a quadratic function of the states ($g - \frac{g^2}{2}$) or an exponential ($1 - \exp(-g)$). Choosing a basis of only order 2 is by consequence not sufficient. We decided to go up to a degree 4.

We can now plot Q^θ found using the LSTD algorithm against $\hat{Q}$ found by summing the reward over a long trajectory using the points in our dataset as initial conditions:

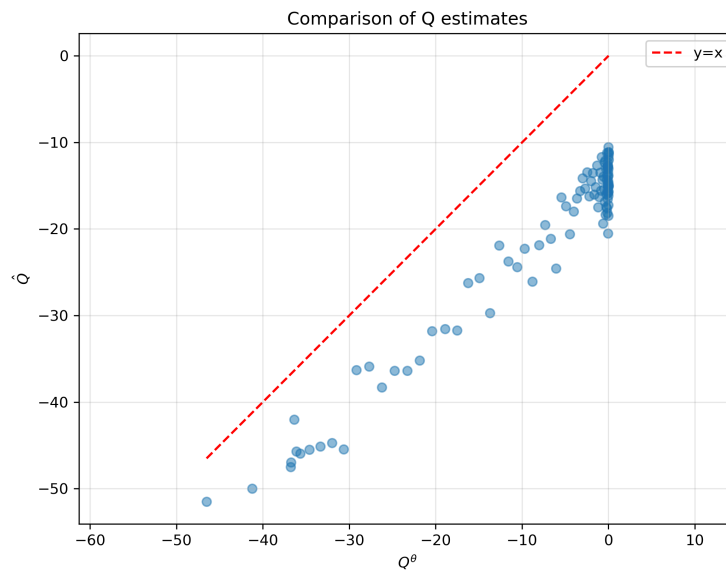


Figure 25: Q^θ against $\hat{Q}$ for every points in our dataset

We can see that there is only a slight shift between Q^θ and $\hat{Q}$. This shift is not a problem as the Q -function satisfies the Bellman equation up to a constant and, in our policy improvement step, we want to maximize it anyway so the constant does not bother us.

Question 6.4

We applied the LSTD algorithm to the LQR approximate model while still using the π_{cl} policy. We thus use this reward:

$$\hat{r}(x_t, u_t) = \frac{1}{2}x^\top Qx + x^\top Su + u^\top Ru$$

where the matrix Q , S , and R have been found in the previous part where we derived the LQR model. For LSTD, we chose an initial condition $x_0 = (1, 1, 1)^\top$ and a variance for the exploration policy of $\sigma_{\text{exp}} = 0.1$ because we saw that LSTD converged with those parameters. For the basis choice, we know that the Q -function in the LQR model has a quadratic form so we chose:

$$\begin{aligned} \psi(x, u) = & \left(q^2, 2qz^a, 2qz^u, 2qu \right. \\ & (z^a)^2, 2z^az^u, 2z^au \\ & \left. (z^u)^2, 2z^uu, u^2 \right) \end{aligned}$$

which gives us a basis with $d = 10$. After solving LSPD and obtaining the optimal parameters θ^{LSTD} , we plotted the LSTD Q -function against the empirical Q -function $\hat{Q}$:

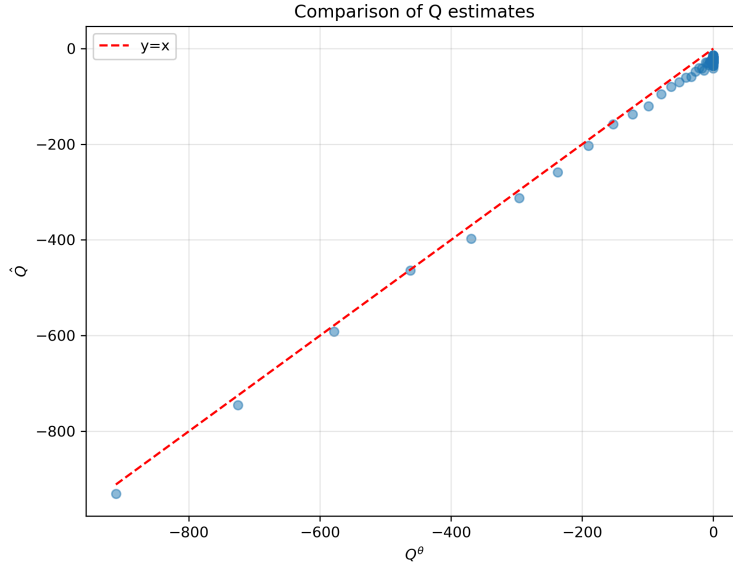


Figure 26: Q^θ against $\hat{Q}$ for every points in our dataset

We can see that both Q -function matches almost perfectly. This is expected as the exact Q -function is quadratic and can thus be exactly fit by a linear combination of ψ . In fact, we know that the Q -function is defined as:

$$Q(x, u) := c(x, u) + J^*(F(x, u))$$

where c is the cost, F is the transition function of the dynamics and J^* is the optimal value function. In the case of the LQR model, we have access to a closed form expression of the optimal value function:

$$J^*(x) = x^\top M^* x$$

where we computed M^* while solving the Ricatti equation to obtain the optimal gains. It is thus possible to obtain an exact expression of the Q -function.

Question 6.5

Now that we have obtained an approximation of the Q -function for the original problem, we can include policy improvement in the algorithm to get LSPI. To get the optimal policy, we solved an optimization problem of the Q -function for the input u and we did that for many samples x . We then fitted a linear policy $u = -Kx$ using all the samples to obtain K . After performing this for a few iterations, we simulated trajectories for π_{cl} , π_{lqr} and π_{lspi} and compared their rewards:

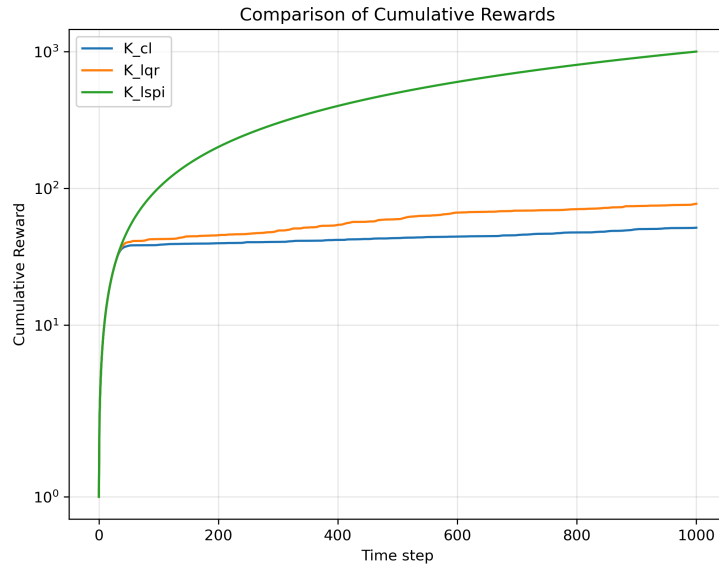


Figure 27: Q^θ against $\hat{Q}$ for every points in our dataset

Here, we see that the LSPI policy performed way better in this model than the other two rewards.

Question 6.6

We implemented LSPI in the settings of question 6.4. and plotted the difference in norm between the policy obtained after k iteration and the optimal LQR policy K_{lqr} for each k . Here are the results:

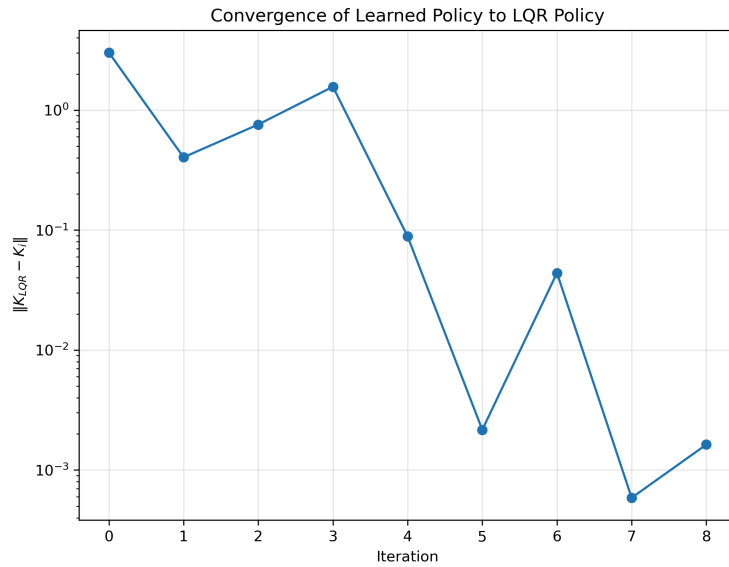


Figure 28: Convergence of the policy learned by LSPI

As we can see, even when we start with K_{cl} as our initial policy, we end up getting very close to our LQR optimal policy.

Question 6.7

We now added the constraint given in the statement during the policy improvement step. We obtain a new policy from LSPI and we plotted the rewards for the different policies that we also clipped for a fair comparison:

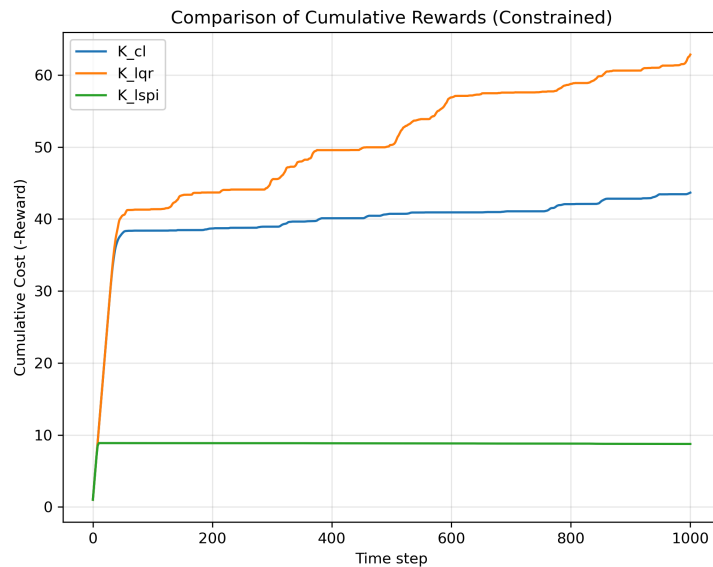
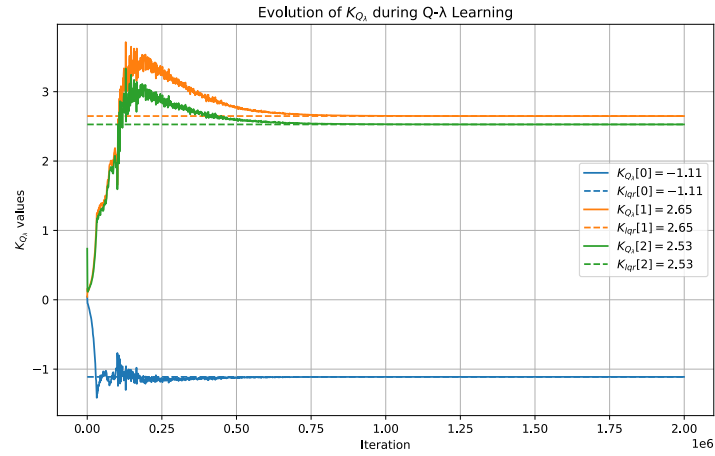


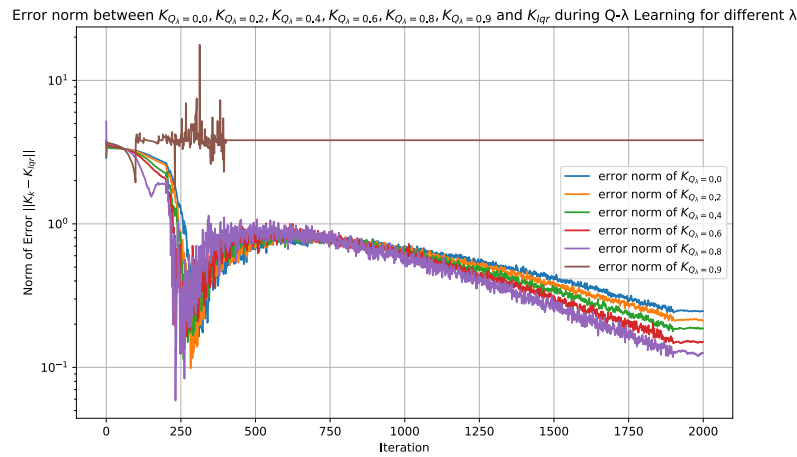
Figure 29: Q^θ against $\hat{Q}$ for every points in our dataset

Here we see that the LSPI policy actually performs worse than the other two.

Question 7.1



Q- λ learning weights are in fact converging to the K_{lqr} gains.



We observe that a greater λ improve the convergence rate.

We also observe that for $\lambda = 0.9$, the gradient explodes and it doesn't converge. This shows that greater λ introduce instability in the system. A way to solve this is to decrease the learning rate α . λ improve the convergence rate but the effect is smoothed over multiple steps so it is better than just increasing α .

Question 7.2

Let $\pi_Q = K_Q^T x$ with $K_Q = [-0.086; 2.1 \cdot 10^{-4}; 0.012]$ trained with Q- λ learning on the deterministic true system.

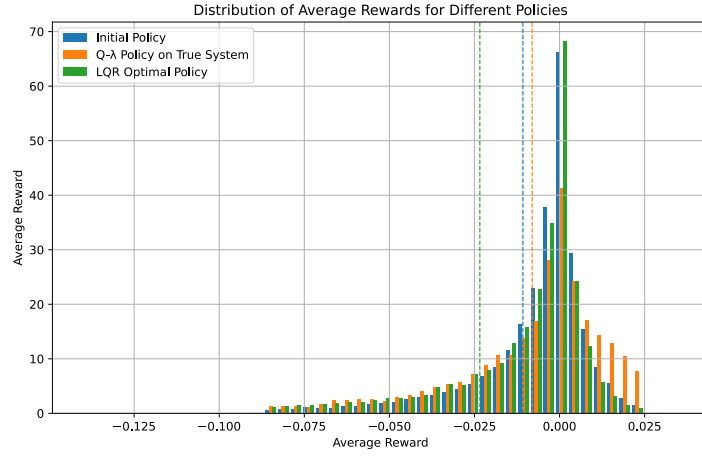
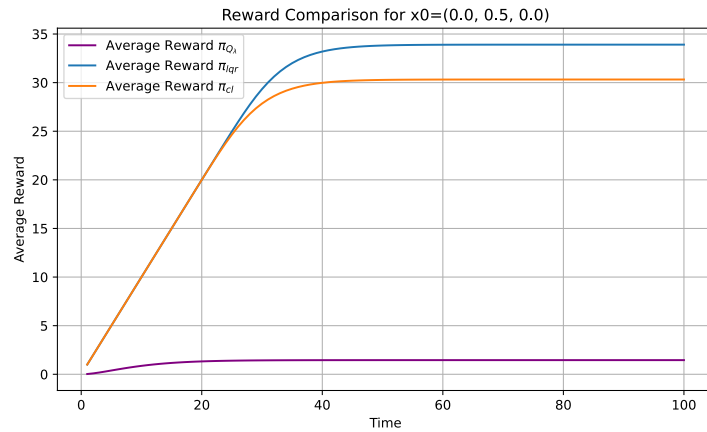
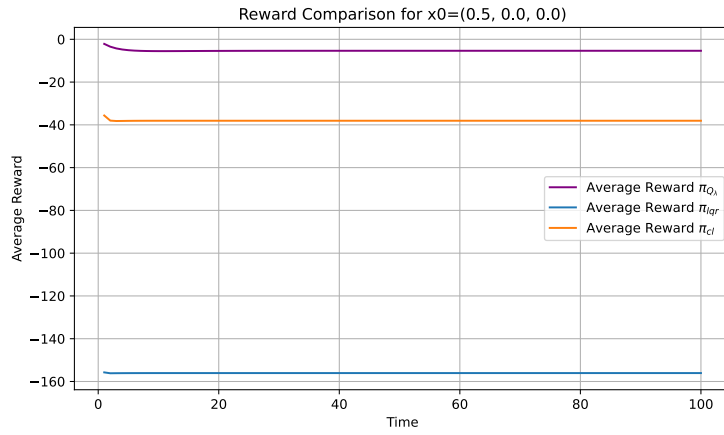
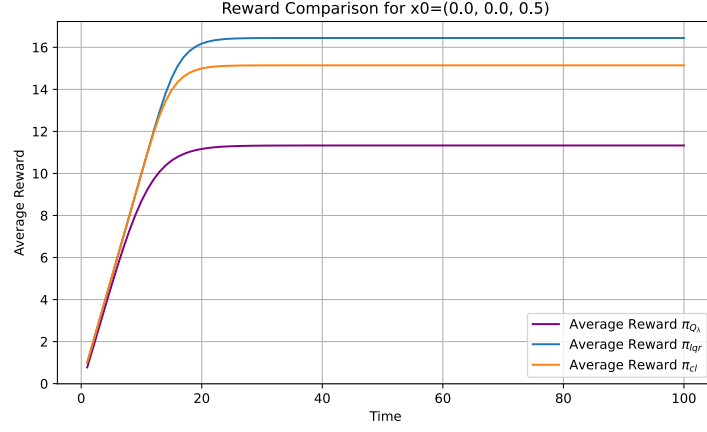


Figure 32: average reward of $\pi_{cl}, \pi_{lqr}, \pi_Q$ over the deterministic system.

It is in fact performing better than Q- λ on the approximated system and than π_{cl} but only on the deterministic system. Only initial condition where random for the graph above.





We observe that π_Q is more cautious, making less gains but also less losses. This makes sense because the true system reward is more for having lower variance. We also see that the gains occur at the start of the episode and then there are no more gains. This is because there is no random variation in the model and the closed loop model is stabilizing itself to $(0, 0, 0)$. That also shows why we used $T = 50$ for the training of this policy.

References

- [1] “Linear–quadratic regulator.” [Online]. Available: https://en.wikipedia.org/wiki/Linear%E2%80%93quadratic_regulator#Infinite-horizon,_discrete-time
- [2] Sean Meyn, *Control Systems & Reinforcement Learning*. 2025.